

Q1.

The table shows some of the outcomes of performing a modular arithmetic operation.

	2	3
2		1
3	1	

Which pair are operations that could each be represented by the table?

Tick (✓) **one** box.

☐

Addition mod 6 and multiplication mod 5

☐

Addition mod 6 and multiplication mod 6

☐

Addition mod 4 and multiplication mod 5

☐

Addition mod 4 and multiplication mod 6

(Total 1 mark)

Q2.

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The binary operation $\otimes$ is given by

$$a \otimes b = 3a(5 + b) \pmod{8}$$

where $a, b \in \mathbb{Z}$

Given that $2 \otimes x = 6$, which of the integers below is a possible value of x ?

Circle your answer.

0 1 2 3

(Total 1 mark)